

TW1.1 - Git Workflow & Collaboration Report

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Subject: DevOps Lab

Aim

To understand distributed version control using Git and GitHub and perform repository creation, cloning, branching, committing, merging and pushing changes.

Objectives

- Learn Git basics.
- Work with local and remote repositories.
- Understand branching and merging.
- Collaborate using GitHub.

Theory

Git is a distributed version control system developed by Linus Torvalds. It allows developers to track changes in source code, maintain version history, collaborate efficiently, resolve merge conflicts, and manage software projects. GitHub is a cloud platform that hosts Git repositories and enables collaboration through pull requests, issues, and repository management.

Software Requirements

Git, GitHub account, Windows/Linux, Internet connection, VS Code.

Procedure

The screenshot displays the VS Code interface during a Git merge conflict resolution. The Source Control panel on the left shows the 'feature/user-auth' branch being merged into 'main'. The code editor in the center shows the conflict between the 'Incoming' branch (feature/user-auth) and the 'Current' branch (main). The 'Incoming' view shows the code from 'feature/user-auth' with a yellow box highlighting the conflict resolution options: 'Accept Incoming | Accept Combination | Ignore'. The 'Current' view shows the code from 'main' with a yellow box highlighting the same options. The 'Result' view shows the code after the merge, with a yellow box indicating 'No Changes Accepted' for the conflicting lines. A 'Complete Merge' button is visible at the bottom right.

```
Assignment-TW1.1-Git > app.py > ...
1 from flask import Flask
2
3 app = Flask(__name__)
4
5 @app.route("/")
6 def hello_world():
7     return "Hello, Docker! Main Branch + User Authentication Feature"
8
9 if __name__ == "__main__":
10     app.run(host="0.0.0.0", port=5000)
```

Terminal Output:

```
[feature/user-auth f5f998d] Updated message in feature branch
4 files changed, 1 insertion(+), 1 deletion(-)
rename Screenshots/(Screenshot 2025-07-27 133014.png -> git tw1.1.png) (100%)
create mode 100644 Screenshots/github branch.png
create mode 100644 Screenshots/modification auth.png
PS C:\Users\Adarsh Jha\Desktop\DevOps-Lab-Assignments> git checkout main
Switched to branch 'main'
Your branch is based on 'origin/main', but the upstream is gone.
(use "git branch --set-upstream" to fixup)
PS C:\Users\Adarsh Jha\Desktop\DevOps-Lab-Assignments> git add .
PS C:\Users\Adarsh Jha\Desktop\DevOps-Lab-Assignments> git commit -m "Updated message in main branch"
[main 5b8dc93] Updated message in main branch
1 file changed, 1 insertion(+), 1 deletion(-)
PS C:\Users\Adarsh Jha\Desktop\DevOps-Lab-Assignments> git merge feature/user-auth
Auto-merging Assignment-TW1.1-Git/app.py
CONFLICT (content): Merge conflict in Assignment-TW1.1-Git/app.py
Automatic merge failed; fix conflicts and then commit the result.
```

Incoming ϕ f5f998d • refs/heads/feature/user-auth ✓

Current ϕ 5b8dc93 • refs/heads/main ✓

Result Assignment-TW1.1-Git/app.py 1 Conflict Remaining

```
5 @app.route("/")
6 def hello_world():
7     return "Hello, Docker!"
8
9 if __name__ == "__main__":
10     app.run(host="0.0.0.0", port=5000)
```

Complete Merge

app.py 5.1 X

Assignment-TW1.1-Git > app.py > ...

```
1 from flask import Flask
2
3 app = Flask(__name__)
4
5 @app.route("/")
6 def hello_world():
7     return "Hello, Docker! Main Branch"
8
9
10 return "Hello, Docker! User Authentication is working!"
11
12 >>>>>> feature/user-auth (incoming change)
13
14 if __name__ == "__main__":
15     app.run(host="0.0.0.0", port=5000)
```

Resolve in Merge Editor

OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS GITLENS AZURE

• [feature/user-auth f5f98bd] Updated message in feature branch
4 files changed, 1 insertion(+), 1 deletion(-)
rename Screenshots/(Screenshot 2026-07-27 133014.png -> git twi.1.png) (100%)
create mode 100644 Screenshots/github branch.png
create mode 100644 Screenshots/modification auth.png
• PS C:\Users\Adarsh Jha\Desktop\DevOps-Lab-Assignments> git checkout main
Switched to branch 'main'
Your branch is based on 'origin/main', but the upstream is gone.
(use "git branch --unset-upstream" to fixup)
• PS C:\Users\Adarsh Jha\Desktop\DevOps-Lab-Assignments> git add .
PS C:\Users\Adarsh Jha\Desktop\DevOps-Lab-Assignments> git commit -m "Updated message in main branch"
[main 5b8dc93] Updated message in main branch
1 file changed, 1 insertion(+), 1 deletion(-)
• PS C:\Users\Adarsh Jha\Desktop\DevOps-Lab-Assignments> git merge feature/user-auth
Auto-merging Assignment-TW1.1-Git/app.py
CONFLICT (content): Merge conflict in Assignment-TW1.1-Git/app.py
Automatic merge failed; fix conflicts and then commit the result.
• PS C:\Users\Adarsh Jha\Desktop\DevOps-Lab-Assignments>

main+1 niccokunzmann/first_timer_scraper#28 needs reviewers

Ln 14, Col 39 Spaces: 4 UTF-8 CRLF Python Python 3.14 (64-bit) Go Live Pylint: 5.00/10

EXPLORER

DEVOPS-LAB-ASSIGN... Assignment-TW1.1-Git

Assignment-TW1.1-Git

Assignment-TW1.2-Jira

Assignment-TW1.3Docker-Jenkins

Project-1-Dockerizing-Jenkins-Pipeli...

Project-2-React-Docker

Project-Distributed-Jenkins

README.md

Screenshots

app.py

```
1 from flask import Flask
2
3 app = Flask(__name__)
4
5 @app.route("/")
6 def hello_world():
7     return "Hello, Docker!"
8
9 if __name__ == "__main__":
10     app.run(host="0.0.0.0", port=5000)
```

Assignment-TW1.1-Git > app.py You, 28 seconds ago via PR ... Initial Flask Hello World Application ...

OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS GITLENS AZURE

PS C:\Users\Adarsh Jha\Desktop\DevOps-Lab-Assignments> git status

No commits yet

Changes to be committed:
(use "git rm --cached <file>..." to unstage)
new file: Assignment-TW1.1-Git/Dockerfile
new file: Assignment-TW1.1-Git/app.py
new file: Assignment-TW1.1-Git/requirements.txt

PS C:\Users\Adarsh Jha\Desktop\DevOps-Lab-Assignments> git commit -m "Initial Flask Hello World Application"

[main (root-commit) 9973cdd] Initial Flask Hello World Application
3 files changed, 25 insertions(+)
create mode 100644 Assignment-TW1.1-Git/Dockerfile
create mode 100644 Assignment-TW1.1-Git/app.py
create mode 100644 Assignment-TW1.1-Git/requirements.txt

• PS C:\Users\Adarsh Jha\Desktop\DevOps-Lab-Assignments> git log --oneline
9973cdd (HEAD -> main) Initial Flask Hello World Application
• PS C:\Users\Adarsh Jha\Desktop\DevOps-Lab-Assignments>

main+1 niccokunzmann/first_timer_scraper#28 needs reviewers

Ln 1, Col 1 Spaces: 4 UTF-8 CRLF Python

AdarshCodes1221 / DevOps-Lab-Assignments

+

Code

Issues

Pull requests

Agents

Actions

Projects

Wiki

More

Watch 0

The entire DevOps Lab from start to finish.

0 stars

0 forks

0 watching

1 Branch

0 Tags

Activity

Public repository

feature/user-auth

Go to file

+

Code

AdarshCodes1221

Added user authentication feature

c0e1e33 · 2 minutes ago

Assignment-TW1.1-Git

Added user authentication feature

2 minutes ago

Screenshots

Added user authentication feature

2 minutes ago

File

Edit

Selection

View

Go

Run

Terminal

Help

my-flask-app

Welcome

app.py

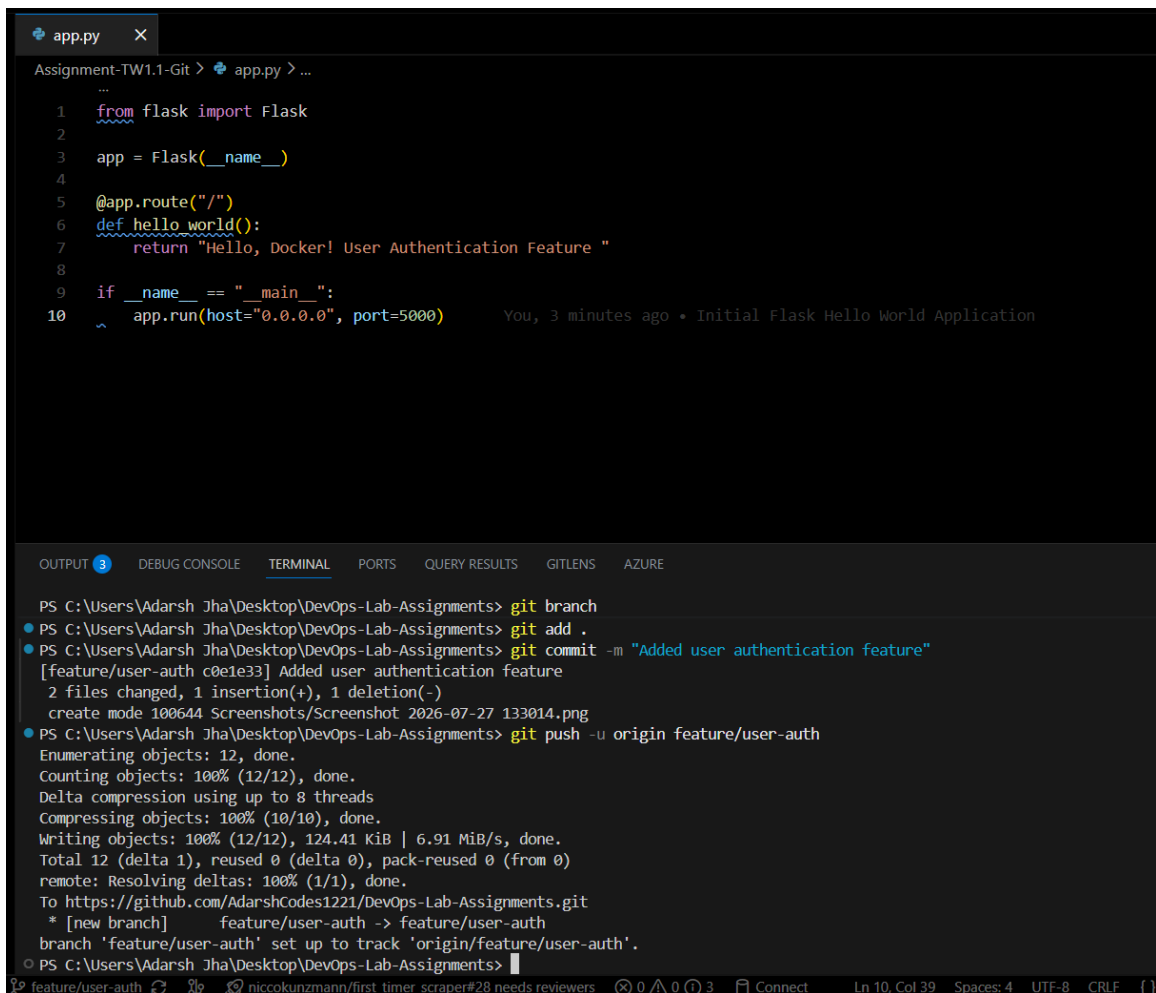
Dockerfile

requirements.txt

127.0.0.1:5000

http://127.0.0.1:5000/

Hello, Docker!



The image shows a VS Code editor window with a file named `app.py` open. The code is a simple Flask application that returns "Hello, Docker! User Authentication Feature " when accessed at the root URL. The terminal window at the bottom shows the following commands and output:

```
PS C:\Users\Adarsh Jha\Desktop\DevOps-Lab-Assignments> git branch
PS C:\Users\Adarsh Jha\Desktop\DevOps-Lab-Assignments> git add .
PS C:\Users\Adarsh Jha\Desktop\DevOps-Lab-Assignments> git commit -m "Added user authentication feature"
[feature/user-auth c0e1e33] Added user authentication feature
2 files changed, 1 insertion(+), 1 deletion(-)
create mode 100644 Screenshots/Screenshot 2026-07-27 133014.png
PS C:\Users\Adarsh Jha\Desktop\DevOps-Lab-Assignments> git push -u origin feature/user-auth
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 8 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (12/12), 124.41 KiB | 6.91 MiB/s, done.
Total 12 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/AdarshCodes1221/DevOps-Lab-Assignments.git
 * [new branch]      feature/user-auth -> feature/user-auth
branch 'feature/user-auth' set up to track 'origin/feature/user-auth'.
PS C:\Users\Adarsh Jha\Desktop\DevOps-Lab-Assignments>
```

Commands Used

git init, git clone, git status, git add ., git commit -m, git branch, git checkout, git merge, git push, git pull.

Learning Outcomes

Learned version control, repository management, collaborative development, and basic Git workflow.

Conclusion

Git simplifies version control and enables efficient collaboration while maintaining complete project history.